

BASHAIR YAQOOB

Junior AI / Software Engineer Candidate
Karachi, Pakistan

GitHub: <https://github.com/BashairYaqoob>

SUMMARY

Computer Science student and software development enthusiast with hands-on experience building web applications, AI-powered projects, and programming projects. Interested in AI engineering, LLM applications, retrieval-augmented generation, and practical software development.

Experienced with Python, JavaScript, TypeScript, C++, Git, REST APIs, LangChain, and AI application development through academic and personal projects.

Strong interest in building complete projects from idea and architecture through implementation, debugging, and deployment.

EDUCATION

Bachelor's / Undergraduate Studies in Computer Science

Currently developing practical software engineering and AI skills through coursework, laboratory projects, and independent projects.

TECHNICAL SKILLS

Programming Languages:

- Python
- JavaScript
- TypeScript
- C++
- C

AI / LLM:

- LangChain
- Retrieval-Augmented Generation (RAG)
- LLM application development
- AI chatbots
- Prompt engineering
- Embeddings and retrieval concepts

Web Development:

- HTML
- CSS
- JavaScript
- TypeScript
- React fundamentals
- REST APIs

Tools:

- Git
- GitHub
- Jupyter Notebook
- VS Code

PROJECT EXPERIENCE

AI-Project---MindHaven
AI Mental Health Web Application

GitHub:
<https://github.com/BashairYaqoob/AI-Project---MindHaven>

- Built a mental health-focused web application with an AI chatbot.
- Implemented features for journaling, mood logging, and personal reflection.
- Worked across frontend and backend aspects of the project.
- Used TypeScript as the primary project language.
- Designed the application around an AI-assisted user experience.
- Worked through integration and debugging challenges while developing the application.
- Explained the project architecture, functionality, and development decisions as part of the project.

Whatsapp-Langchain-Bot
WhatsApp AI / RAG Application

GitHub:
<https://github.com/BashairYaqoob/Whatsapp-Langchain-Bot>

- Built a functional WhatsApp bot integrating Baileys and LangChain.
- Implemented a Retrieval-Augmented Generation workflow.
- Worked with information extraction and persistence.
- Used LangChain for the LLM application layer.
- Integrated messaging functionality with an AI workflow.
- Implemented functionality for extracting and persisting personal information.
- Used JavaScript for the project.

Lead-Generation-Agent
AI Lead Generation Agent

GitHub:
<https://github.com/BashairYaqoob/Lead-Generation-Agent>

- Built a lead-generation agent using Python.
- Implemented browser automation for collecting information.
- Added Excel export functionality for generated lead data.
- Worked on connecting multiple components into an end-to-end automation workflow.
- Used Python for the main implementation.

AI
Artificial Intelligence Laboratory Projects

GitHub:
<https://github.com/BashairYaqoob/AI>

- Maintained a collection of AI laboratory work and learning projects.
- Used Jupyter Notebooks to experiment with artificial intelligence concepts.
- Developed practical familiarity with AI concepts through experimentation and implementation.

Pacman-Game Object-Oriented Programming Project

GitHub:
<https://github.com/BashairYaqoob/Pacman-Game>

- Built a Pacman game as an object-oriented programming project.
- Implemented the project using C++.
- Practiced object-oriented design, program structure, and debugging.

ADDITIONAL EXPERIENCE

- Built academic and personal software projects across AI, web development, automation, and programming.
- Comfortable working across frontend and backend components.
- Regularly use Git and GitHub for project development.
- Learn new technologies through hands-on implementation and experimentation.
- Comfortable working independently as well as in a team environment.

RELEVANT EXPERIENCE FOR THIS ROLE

Python:
Hands-on Python experience through AI, automation, laboratory, and software projects.

LLM Frameworks:
Practical experience with LangChain through the WhatsApp LangChain Bot project.

RAG:
Implemented a RAG-based workflow in the WhatsApp LangChain Bot project.

Git / GitHub:
Multiple public software projects maintained on GitHub using Git-based development workflows.

REST / API Integration:
Experience integrating application components and external services within AI and web projects.

End-to-End Project Development:
Experience taking projects from an initial idea through implementation, integration, debugging, and a working project.

AREAS FOR FURTHER DEVELOPMENT

- Deeper understanding of embedding models and vector databases.
- More advanced retrieval-quality evaluation.
- Production-scale RAG architecture.
- Advanced LangGraph workflows.
- Production deployment and observability for AI systems.

GITHUB PROFILE

<https://github.com/BashairYaqoob>